

B. Tech. (3rd Sem)

(Common for CSE, CSE with Specialization in Data Science, CSE with specialization in Cloud Technology & Information Security, CSE with specialization in Big Data and Analytics, CSE with specialization in Full Stack Development, CSE with specialization in AI & Machine Learning)

BCSE-503L (Database Management System Lab)

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Continuous evaluation 60
End semester exam 40
Total marks 100
Credits 1.0

Course Objectives:

- i) Creating and Altering Tables with necessary constraints, keys and data types.
- ii) Inserting data and manipulating data as per needs.
- iii) Writing MySQL Queries to retrieve required information from single/multiple tables.
- iv) Implementing Operations on relations (tables) using MySQL.

List of Practical

1. To Draw ER Model for Library DBMS.
2. To draw Relational model corresponding to ER Model.
3. To Create Tables for Library DBMS.
4. To Describe the Structure of Tables of Library DBMS.
5. To Insert Data In the Tables of Library DBMS.
6. To Select Data From Library Database Using Various Operators.
7. To Describe MySQL Operation including Group By Clause and Use of Sub Queries.
8. To Describe Various MySQL Functions.
9. To Describe MySQL Conversion Functions.
10. To Apply Privileges and Referential Integrity Constraints.
11. To implement various joins in MySql.
12. To implement Regular Expression for searching in MySql.

Projects:

- 1) Consider the company database as shown in Figure 1, which keeps track of employees, departments, and projects.
 - The company is organized into departments. Each department has a unique name, a unique number, and a particular employee who manages the department. A department may have several locations.
 - A department controls a number of projects; each of which has a unique name, a unique number, and a single location.
 - We store each employee's name, emp_id, address, salary, sex, and birth_date. An employee is assigned to one department, but may work on several projects, which are not necessarily controlled by the same department. We also keep track of the direct supervision of each employee.
 - We want to keep track of the dependants of each employee for insurance purpose. We keep each dependant's name, sex, birth_date, and relationship to the employee.

EMPLOYEE

Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	Sex	Salary	Super_ssn	Dno
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DEPARTMENT

Dname	<u>Dnumber</u>	Mgr_ssn	Mgr_start_date
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DEPT_LOCATIONS

<u>Dnumber</u>	<u>Dlocation</u>
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PROJECT

Pname	<u>Pnumber</u>	Plocation	Dnum
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WORKS_ON

<u>Essn</u>	<u>Pno</u>	Hours
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DEPENDENT

<u>Essn</u>	<u>Dependent_name</u>	Sex	Bdate	Relationship
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- 2) Consider the AIRLINE relational database schema, which describes a database for airline flight information as shown in Figure 2. Each FLIGHT is identified by a Flight_number, and consists of one or more FLIGHT_LEGS with Leg_numbers 1, 2, 3, and so on. Each FLIGHT_LEG has scheduled arrival and departure times, airports, and one or more LEG_INSTANCES—one for each Date on which the flight travels. FAREs are kept for each FLIGHT. For each FLIGHT_LEG instance, SEAT_RESERVATIONS are kept, as are the AIRPLANE used on the leg and the actual arrival and departure times and airports. An AIRPLANE is identified by an Airplane_id and is of a particular AIRPLANE_TYPE. CAN_LAND relates AIRPLANE_TYPES to the AIRPORTs at which they can land. An AIRPORT is identified by an Airport_code. Consider an update for the AIRLINE database to enter a reservation on a particular flight or flight leg on a given date.

- Give the operations for this update.
- What types of constraints would you expect to check?
- Which of these constraints are key, entity integrity, and referential integrity constraints, and which are not?
- Specify all the referential integrity constraints that hold on the schema.

AIRPORT

<u>Airport_code</u>	Name	City	State
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FLIGHT

<u>Flight_number</u>	Airline	Weekdays
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FLIGHT_LEG

<u>Flight_number</u>	<u>Leg_number</u>	Departure_airport_code	Scheduled_departure_time
		Arrival_airport_code	Scheduled_arrival_time

LEG_INSTANCE

<u>Flight_number</u>	<u>Leg_number</u>	<u>Date</u>	Number_of_available_seats	Airplane_id
			Departure_airport_code	Departure_time
			Arrival_airport_code	Arrival_time

FARE

<u>Flight_number</u>	<u>Fare_code</u>	Amount	Restrictions
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AIRPLANE_TYPE

<u>Airplane_type_name</u>	Max_seats	Company
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CAN_LAND

<u>Airplane_type_name</u>	<u>Airport_code</u>
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AIRPLANE

<u>Airplane_id</u>	Total_number_of_seats	Airplane_type
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SEAT_RESERVATION

<u>Flight_number</u>	<u>Leg_number</u>	<u>Date</u>	<u>Seat_number</u>	Customer_name	Customer_phone
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Course Outcomes: After completing this course, students will be able to:

- Design a Database without anomalies as per requirements.
- Construct complex queries to retrieve required information from database.
- To implement various integrity constraints.
- Creating views and manipulating them as needed.